

HCSM-00

Foundations, Notation, and Mapping

The Hodge Complex Standard Model: Substrate, Hodge Complex, Spectral Midpoint,
Processor, Output Structures, Emergent Physics, and Cosmology

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

September 2026

Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, mixing matrices, spacetime dimension, graviton, Newton constant, and vacuum stability coefficient of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The substrate is selected by two independent anchors (photon helicity for $d = 4$; equal entropy spacing for $L = 8$) and one derived anchor (the topology of the Hodge–Dirac kernel). The Hodge complex of differential forms carries the operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation.

This paper establishes the framework’s foundational objects and states the notation. All seven rank blockers of the framework’s derivation program are now closed: the mass-map coefficient $(d - 1)/2$ from the DNLS Hartree orbit trace; the Cabibbo angle $\pi/14$ from the winding structure; the loop-pass duration from Landauer selection; the kL correction coefficient $(|D_4| - 1)/2$; the anti-daughter coefficients from D_4 orbit sizes; the generation phase $(|F| - |B|)/|F| \cdot \pi$; and the DME entropy-well normalization S_0 .

The framework has three discrete disclosed inputs (Standard Model content; physical identifications; identification thesis) and one dimensionful anchor (the Planck mass M_P). Every other quantity is derived from the substrate. This includes the top-Yukawa coefficient $4/3 = |\mathcal{O}_2|/(d - 1)$; the tree-level quartic coupling $\lambda_{\text{tree}} = \ln(kL)/[(d - 1)\pi^2]$; the bridge identity $\lambda_{\text{tree}} = (\gamma^2/d)(1 - \delta_{\text{edge}}/[2(d - 1)])$; the Cabibbo angle $\sin \theta_{12} = \pi/14$; the neutrino $N/4$ spectrum (60, 60, 58); the mass-map coefficient $(d - 1)/2 = 3/2$; the loop-pass duration $T_{\text{pass}} = 1/(2\gamma)$; the Newton constant $G_N = (d + 1)^2/(8\pi M_P^2)$; the DME entropy-well normalization $S_0 = v_{\text{EW}}^2 \cdot (2d - 1)/(|F| \cdot |B|) \cdot (1 + \delta_{\text{edge}}/(|F| - |B|))$; and the Majorana phases $(\pi, 0)$.

HCSM has fewer empirical anchors than the Standard Model, not that it has none.

Contents

1	Introduction	3
1.1	What HCSM is	3
1.2	The single input $d = 4$	3
1.3	Architectural levels	3
1.4	Scope and honesty	4
1.5	Continuum limit and emergent Lorentz invariance	4

2	Level 0 — The substrate	5
2.1	The periodic 8×8 torus	5
2.2	The discrete 5-point Laplacian	5
2.3	The two independent anchors and the derived third anchor	5
2.4	The size relation $N = d \cdot 2^d$	6
2.5	The 14-mode count at the self-paired eigenvalue	6
3	Level 1 — The Hodge complex	6
3.1	Differential forms on the torus	6
3.2	Exterior derivative and codifferential	6
3.3	The Hodge–Dirac operator	7
3.4	The $\mathbb{Z}_2$ grading	7
3.5	The kernel and the emergence of four dimensions	7
4	Level 2 — The spectral midpoint	7
4.1	The self-paired eigenvalue	7
4.2	The three D_4 orbits	7
4.3	The Hodge lift to the 56-dimensional sector	8
5	Level 3 — The processor and its dynamics	8
5.1	The winding generator	8
5.2	The three winding classes	8
5.3	The processor T — operator-algebraic content	8
5.4	F/B dimensions and internal ratios	9
5.5	The Landauer channel	9
5.6	The zero state	9
5.7	The mirror and diagonal-flip maps	10
5.8	The five-layer loop	10
5.9	The exact ratio and the loop-pass duration	10
6	Level 4 — Output structures	11
6.1	Charge	11
6.2	Mass	11
7	Level 5 — Emergent physics	11
7.1	Flavor: the CKM matrix	11
7.2	Flavor: the PMNS matrix	12
7.3	Higgs sector	12
7.4	Gravity	13
8	Level 6 — Cosmology	13
8.1	The kL crossover	13
8.2	Dark energy and the DME entropy-well normalization	13
8.3	Vacuum stability	14
9	Session-Level Theorem Inventory	14
10	Reproducibility	15

1 Introduction

1.1 What HCSM is

The Hodge Complex Standard Model is a structural derivation of the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework's central claim is that the Standard Model is not a fundamental description of nature, but the residue left behind when a small, discrete, information-bearing lattice cancels itself against its own mirror.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

HCSM does not ask what fields exist or what symmetry groups are broken. It asks a single question: what substrate could produce exactly the Standard Model, and nothing else? The framework's answer is a discrete lattice whose mirror cancellation leaves exactly the observed particle content.

1.2 The single input $d = 4$

The whole structure of HCSM traces back to a single input: the spacetime dimension $d = 4$. This input is itself derived from the photon's two helicity states, and the substrate size $L = 8$ is selected independently by the equal-entropy-spacing theorem on the torus Laplacian. The two derivations agree, and together they force the substrate and the emergent dimension.

1.3 Architectural levels

The framework's architecture is a chain from the substrate to observable physics, organized in six levels.

Level 0. Substrate. The periodic 8×8 torus Λ , the discrete 5-point Laplacian, the two independent anchors.

Level 1. Hodge complex. The differential forms $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$, the Hodge–Dirac operator $D = d + d^*$, the $\mathbb{Z}_2$ grading, the kernel $\ker D \cong H^0 \oplus H^1 \oplus H^2$.

Level 2. Spectral midpoint. The self-paired eigenvalue $\lambda = 4$, its multiplicity 14, the three D_4 orbits, the Hodge lift to the 56-dimensional sector.

Level 3. Processor and dynamics. The winding generator H_{wind} , the three winding classes, the processor $T = I - P_C$, the fermion and boson sectors, the Landauer channel, the loop operator.

Level 4. Output structures. The charge functional q_3 , the mass map, the $N/4$ labels.

Level 5. Emergent physics. Flavor (CKM, PMNS), the Higgs sector, gravity, the Newton constant.

Level 6. Cosmology. The kL crossover, dark energy, the DME entropy-well normalization, vacuum stability.

1.4 Scope and honesty

Every theorem stated as HCSM-native uses only the following primitives:

- the periodic 8×8 torus Λ ;
- the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$;
- the Hodge–Dirac operator $D = d + d^*$;
- the D_4 point-group action on Λ ;
- the discrete 5-point Laplacian Δ ;
- the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$;
- the zero-state vector s ;
- the charge functional $q_3 = F(n_1, \sigma)$;
- the $N/4$ labels;
- the processor $T = P_{\text{mesh}} + P_7$ and its class partition.

No imported machinery appears in any HCSM-native theorem. Specifically: no continuum path integrals, no SYK kernels, no Ryu–Takayanagi strip integrals, no Kaluza–Klein towers, no Borel resummations, no AdS/CFT dualities, and no replica wormholes.

The framework’s disclosed-input set consists of three discrete items (Standard Model per-generation content; physical identifications $A_1 \leftrightarrow \text{color}$, $B_1 \leftrightarrow \text{weak isospin}$; identification thesis 14 modes $\leftrightarrow$ 14 SM particles) and one dimensionful anchor (the Planck mass M_P). The single-anchor structure is established by HCSM-53 §5: the DME entropy-well normalization S_0 is derived from M_P and HCSM-native dimensionless factors. No second dimensionful anchor is required.

1.5 Continuum limit and emergent Lorentz invariance

The substrate is not a discretization of an underlying continuum. The framework’s claim is the opposite: the continuum emerges from the substrate. This subsection clarifies the sense in which the finite-dimensional lattice content of HCSM produces an infinite-dimensional quantum field theory in the physical scaling limit.

Physical vs. lattice scale. The lattice spacing is not a physical length; it is a structural count. The physical scale is set by the mass map and the kL crossover (Levels 4 and 6). The dimensionless ratio

$$\epsilon = \frac{\text{lattice scale}}{\text{physical scale}} \quad (1)$$

is the small parameter of the scaling limit. Physical observables of HCSM are functions of ϵ that remain finite as $\epsilon \rightarrow 0$.

Dispersion and isotropy. The discrete 5-point Laplacian has the expansion

$$\lambda(n_1, n_2) = 4 - 2 \cos \frac{\pi n_1}{4} - 2 \cos \frac{\pi n_2}{4} = \frac{\pi^2}{16} (n_1^2 + n_2^2) + O(n^4), \quad (2)$$

isotropic to leading order. The quartic corrections are of order $\pi^4(n_1^4 + n_2^4)/768$; they are suppressed by ϵ^2 relative to the leading term.

Cohomological origin of the physical content. The physical content of HCSM does not live in the individual lattice modes but in the cohomology of the Hodge complex. The kernel $\ker D \cong H^0(T^2) \oplus H^1(T^2) \oplus H^2(T^2)$ is a topological invariant, independent of the lattice spacing. The four harmonic forms give the four spacetime directions; the Lorentzian signature $(3, 1)$ is fixed by the Hodge star on $\ker D$, not by the lattice geometry.

Lorentz invariance. The $SO(3, 1)$ Lorentz symmetry is emergent, not imposed. It acts on the kernel $\ker D$ via the intersection form on $H^0 \oplus H^2$ and the Hodge inner product on H^1 (HCSM-03). The group of isometries of the signature- $(3, 1)$ metric on the four-dimensional kernel is exactly $SO(3, 1)$, with no lattice remnant.

2 Level 0 — The substrate

2.1 The periodic 8×8 torus

The substrate of HCSM is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad (3)$$

consisting of $N = 64$ sites arranged in an 8×8 grid with periodic boundary conditions. Site (x, y) is identified with $(x + 8, y)$ and with $(x, y + 8)$.

Topologically, the periodic lattice is equivalent to the torus $T^2 = S^1 \times S^1$.

2.2 The discrete 5-point Laplacian

Each site (x, y) has four nearest neighbors: $(x \pm 1, y)$ and $(x, y \pm 1)$, with all coordinates modulo 8. The natural differential operator on the lattice is the discrete 5-point Laplacian

$$(\Delta f)(x, y) = 4f(x, y) - f(x + 1, y) - f(x - 1, y) - f(x, y + 1) - f(x, y - 1). \quad (4)$$

The eigenvalues of Δ on plane waves $\phi_{n_1, n_2}(x, y) = e^{2\pi i(n_1 x + n_2 y)/8}$ are

$$\lambda(n_1, n_2) = 4 - 2\cos\left(\frac{\pi n_1}{4}\right) - 2\cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (5)$$

2.3 The two independent anchors and the derived third anchor

The size $L = 8$ is not put in by hand. It is determined by two independent anchors. A third anchor follows as a theorem.

Anchor 1 (input): photon helicity. A massless spin-1 particle has exactly two helicity states. Each is described by a complex amplitude with two real components. Two states $\times$ two components = four real numbers, spanning a four-dimensional phase space whose Wick rotation produces four-dimensional Lorentzian spacetime. Hence $d = 4$.

Anchor 2 (input): equal entropy spacing. For an $L \times L$ torus, each mode has a Shannon entropy in bits. The modes fall into entropy classes. Among all integers $4 \leq L \leq 10^4$, the only L for which the torus Laplacian has exactly three equally spaced entropy classes is $L = 8$. The three classes are 6.0, 5.5, 5.0 bits, with spacing 0.5 bits. Hence $L = 8$.

Anchor 3 (derived): topology of the kernel. Given the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ selected by Anchors 1 and 2, the kernel of the Hodge–Dirac operator D on Λ has dimension $b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$, where (b_0, b_1, b_2) are the Betti numbers of T^2 . This is stated and proved in §3.5 as Theorem 4. It is not an independent input.

The logical dependency graph is:

$$\text{Anchor 1} \rightarrow d = 4, \quad \text{Anchor 2} \rightarrow L = 8, \quad \{d = 4, L = 8\} \rightarrow N = d \cdot 2^d = 64 \rightarrow \text{Anchor 3}. \quad (6)$$

2.4 The size relation $N = d \cdot 2^d$

Theorem 1 (Substrate size). *The number of sites satisfies $N = d \cdot 2^d$.*

Proof. The photon is a 1-form with d independent components. The Hodge complex in d dimensions has 2^d independent form-degree sectors. The substrate is their tensor product, giving $N = d \cdot 2^d$. At $d = 4$, $N = 64$, a perfect square; this forces $L = \sqrt{N} = 8$. Among $1 \leq d \leq 8$, only $d = 4$ makes $d \cdot 2^d$ a perfect square. $\square$

2.5 The 14-mode count at the self-paired eigenvalue

The torus has exactly 14 modes at the self-paired eigenvalue $\lambda = 4$:

$$\text{mult}_{\lambda=4} = 2(L - 1) = 2(8 - 1) = 14. \quad (7)$$

This matches one Standard Model generation. For any other L , the count is wrong.

3 Level 1 — The Hodge complex

3.1 Differential forms on the torus

On the two-dimensional torus Λ , there are three types of differential forms:

- Ω^0 : 0-forms (functions on the 64 sites), dimension 64, bosonic;
- Ω^1 : 1-forms (fields on the 128 directed edges), dimension 128, fermionic;
- Ω^2 : 2-forms (fields on the 64 plaquettes), dimension 64, bosonic.

The total dimension is $64 + 128 + 64 = 256$.

3.2 Exterior derivative and codifferential

For a 0-form f and a 1-form $\omega = (\omega_1, \omega_2)$, the exterior derivatives are

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad (8)$$

$$(d_1 \omega)(x) = \omega_2(x + e_1) - \omega_2(x) - \omega_1(x + e_2) + \omega_1(x). \quad (9)$$

The codifferential $d^* = d^\dagger$ is the adjoint with respect to the standard inner products. They satisfy exact nilpotency: $d_1 d_0 = 0$.

3.3 The Hodge–Dirac operator

Theorem 2 (Hodge–Dirac operator). *The Hodge–Dirac operator is $D = d + d^*$, acting on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.*

Theorem 3 (Hodge–Dirac square). *On the discrete torus Λ , the identity $D^2|_{\Omega^k} = \Delta$, $k = 0, 1, 2$, holds exactly on each form degree, where Δ is the discrete 5-point Laplacian.*

The identity $D^2 = \Delta$ is exact, not approximate. Unlike the spinorial Dirac operator on the same lattice, which suffers from fermion doubling, the Hodge–Dirac square is the ordinary scalar Laplacian.

3.4 The $\mathbb{Z}_2$ grading

Even forms ($\Omega^0 \oplus \Omega^2$) are bosonic; odd forms (Ω^1) are fermionic. The Hodge–Dirac operator is odd with respect to this grading: it is the supercharge. This $\mathbb{Z}_2$ grading is the origin of the spin-statistics connection in HCSM.

3.5 The kernel and the emergence of four dimensions

Theorem 4 (Kernel dimension). *By the discrete Hodge theorem, the kernel of D is isomorphic to the de Rham cohomology of the torus:*

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}), \quad (10)$$

with dimension $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.

The four harmonic forms of Theorem 4 are the four spacetime directions. This is the derivation of Anchor 3 from the substrate selected by Anchors 1 and 2.

The canonical metric on $\ker D$ is constructed from the intersection form $Q(\alpha, \beta) = \int \alpha \wedge \beta$ on $H^0 \oplus H^2$ and the Hodge inner product $\langle \alpha, \beta \rangle = \int \alpha \wedge \star \beta$ on H^1 . The resulting metric has signature $(3, 1)$ with no free coefficients. The timelike direction is the self-dual 0-form/2-form combination $(v_0 + v_2)/\sqrt{2}$.

The group of isometries of this metric is exactly $SO(3, 1)$; Lorentz invariance is therefore an exact emergent symmetry of the kernel, not a lattice artifact.

4 Level 2 — The spectral midpoint

4.1 The self-paired eigenvalue

Solving $\lambda = 8 - \lambda$ in the mirror symmetry of the Laplacian spectrum gives the self-paired midpoint $\lambda = 4$. This eigenvalue has multiplicity 14 and cannot cancel with any mirror partner. The 14 modes at $\lambda = 4$ are the residue that survives.

4.2 The three D_4 orbits

The 14 modes at $\lambda = 4$ decompose under the D_4 point group of the square into three orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (11)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (12)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (13)$$

The support classes are $\{64, 48, 32\} = \{N, V, N/2\}$ with $V = (d-1) \cdot 2^d = 48$. The inverse participation ratios are exactly $2/128, 3/128, 4/128$. The Shannon entropies are exactly 6.0, 5.5, 5.0 bits, with equal spacing 0.5 bits.

4.3 The Hodge lift to the 56-dimensional sector

The 14-fold degeneracy at $\lambda = 4$ lifts to the full Hodge complex:

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (14)$$

This 56-dimensional sector is the arena of the physical world. Its D_4 content is

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (15)$$

5 Level 3 — The processor and its dynamics

5.1 The winding generator

The processor acts on the 14-mode multiplet at $\lambda = 4$. Its first move is the winding generator

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8}M\right), \quad M_{ij} = n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)}, \quad (16)$$

where $(n_1^{(i)}, n_2^{(i)}) \in \mathbb{Z}_8 \times \mathbb{Z}_8$ are the momentum labels of mode i . The matrix H_{wind} is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$.

The characteristic polynomial of H_{wind} restricted to the 14-mode subspace is

$$p(\lambda) = \det(\lambda I - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32). \quad (17)$$

The spectrum is $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$ with multiplicities 1, 1, 10, 1, 1.

5.2 The three winding classes

The 14 modes partition into three classes by frequency:

$$\text{Class A} = \{(0, 4), (4, 0), (6, 6)\}, \quad \omega_A = 2\sqrt{2}, \quad (18)$$

$$\text{Class B} = \{(1, 3), (1, 5), (3, 1), (3, 7), (7, 3), (7, 5), (5, 1)\}, \quad \omega_B = 6\sqrt{2}, \quad (19)$$

$$\text{Class C} = \{(5, 7), (2, 2), (2, 6), (6, 2)\}, \quad \omega_C = 0. \quad (20)$$

The class projectors P_A, P_B, P_C are diagonal in the mode basis, orthogonal, idempotent, and sum to the identity. The ratio $\omega_B/\omega_A = 3 = d-1$ is exact.

5.3 The processor T — operator-algebraic content

The processor T is the rank-10 orthogonal projection

$$T = P_{\text{mesh}} + P_\tau = I - P_{\text{sub}} + P_\tau = I - P_C, \quad (21)$$

where P_{mesh} projects onto the 9 mesh resonances ($\delta \neq 0$), P_τ projects onto the unique mode with zero-state amplitude $s = +1$, and P_C projects onto the winding-zero class. Its image is the fermion sector F , its kernel is the boson sector B :

$$F = \{\delta \neq 0\} \cup \{s = +1\}, \quad |F| = 10, \quad B = \{\delta = 0\} \setminus \{s = +1\}, \quad |B| = 4. \quad (22)$$

Proposition 5 (Operator-algebraic characterization). *T is the unique rank-10 orthogonal projection on the 14-mode multiplet that*

1. *is self-adjoint: $T^\dagger = T$;*
2. *is idempotent: $T^2 = T$;*
3. *is D_4 -equivariant: $[T, g] = 0$ for every $g \in D_4$;*
4. *has kernel equal to the winding-zero class C .*

5.4 F/B dimensions and internal ratios

Theorem 6 (F/B dimensions). *The dimensions of the fermion and boson sectors and their internal ratios are*

$$|F| = \frac{d(d+1)}{2}, \quad |B| = d, \quad \frac{|F| - |B|}{|B|} = \frac{d-1}{2}, \quad \frac{|F|}{|B|} = \frac{d+1}{2}. \quad (23)$$

At $d = 4$: $|F| = 10$, $|B| = 4$, $(|F| - |B|)/|B| = 3/2$, $|F|/|B| = 5/2$.

The quantity $(d-1)/2$ is the mass-map coefficient. The quantity $(d+1)/2$ is the dark-energy coefficient.

5.5 The Landauer channel

The Hodge complex delivers the maximally mixed state on the 56-dimensional sector, with entropy $S_{\text{in}} = \log 56$ nats. The processor reduces this to $S_{\text{out}} = \log 10 + \log 4 = \log 40$. The entropy reduction

$$\Delta S = \log 56 - \log 40 = \log(7/5) = 0.33647 \dots \text{ nats} \quad (24)$$

saturates the deterministic Landauer bound.

5.6 The zero state

A 14×19 measurement matrix M of vacuum-native observables has rank 13 and nullity 1. Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is the zero state. In the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$:

$$s = \begin{pmatrix} -0.105198199128 \\ +0.105198199128 \\ +0.937726561300 \\ -0.192806690737 \\ -0.937726561300 \\ +0.658283218429 \\ -1.000000000000 \\ -0.658283218429 \\ +1.000000000000 \\ +0.192806690737 \\ 0.000000000000 \\ -0.329141609215 \\ +0.329141609215 \\ 0.000000000000 \end{pmatrix}. \quad (25)$$

The vector is odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$. The two fixed points $(2, 2)$ and $(6, 6)$ carry $s = 0$ exactly.

5.7 The mirror and diagonal-flip maps

Definition 7. *On mode labels:*

$$M : (n_1, n_2) \mapsto (4 - n_1, 4 - n_2) \quad (\text{mirror}), \quad (26)$$

$$F : (n_1, n_2) \mapsto (n_2, n_1) \quad (\text{diagonal flip}). \quad (27)$$

Theorem 8 ($\mathcal{O}_2$ self-mirror orbit). *On the 14-mode multiplet at $\lambda = 4$:*

$$|\text{Fix}(M) \cap \mathcal{O}_0| = 0, \quad |\text{Fix}(M) \cap \mathcal{O}_1| = 0, \quad |\text{Fix}(M) \cap \mathcal{O}_2| = 4 = |\mathcal{O}_2|, \quad (28)$$

and

$$\text{Fix}(M) \cap \text{Fix}(F) = \{(2, 2), (6, 6)\}. \quad (29)$$

The orbit $\mathcal{O}_2$ is the unique self-mirror orbit of the 14-mode multiplet. Its two $\mathbb{Z}_2 \times \mathbb{Z}_2$ fixed points are the W boson $(2, 2)$ and the top quark $(6, 6)$.

5.8 The five-layer loop

The loop operator is

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}, \quad (30)$$

acting diagonally on the 14-mode multiplet with polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$. The complex time of mode m is

$$\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m), \quad (31)$$

where $T_{\text{DME}} = 1/(2\gamma)$. The closed forms are

$$\tau_A = 2.0121430664919213 - 0.5521962501528914i, \quad (32)$$

$$\tau_B = 6.0364291994757650 - 1.6508085388210012i, \quad (33)$$

$$\tau_C = 0. \quad (34)$$

5.9 The exact ratio and the loop-pass duration

Theorem 9 (Exact ratio). *The ratio of the two non-trivial loop-eigenvalue magnitudes is $|M_A|/|M_B| = r_A/r_B = 3 = d - 1$.*

Theorem 10 (Loop-pass duration). *The loop-pass duration is*

$$T_{\text{pass}} = T_{\text{DME}} = \frac{1}{2\gamma} = 0.7114000035 \dots \quad (35)$$

Proof. The loop contains the processor $T = I - P_C$, a Maxwell demon at the deterministic Landauer bound. Landauer's principle requires that the entropy reduction be compensated by heat dissipation into the residual bath. Among the HCSM-native scales, only $T_{\text{DME}} = 1/(2\gamma)$ is dissipative; the coherent $T_{\text{beat}} = \pi/\sqrt{2}$ carries no entropy transfer. Because the loop is a measurement, it is irreversible; because it is irreversible, its time unit is the dissipative time. $\square$

6 Level 4 — Output structures

6.1 Charge

The charge theorem states that the electric charge in units of $e/3$ is a bijection

$$q_3 = F(n_1, \sigma), \quad n_1 \in \mathbb{Z}_8, \quad \sigma = \text{sign}(s) \in \{-1, 0, +1\}, \quad (36)$$

from the 14 realized pairs to the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$. The three σ -lines carry charges 0, +5, -8, summing to $-1e$.

6.2 Mass

Theorem 11 (Mass map). *The masses of the twelve charged Standard Model fermions are reproduced by*

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (37)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.5411967342\dots \quad (38)$$

The coefficient $(d-1)/2 = 3/2$ is derived twice. First, from the processor F/B split: $(|F| - |B|)/|B| = (d-1)/2$ (Theorem above). Second, from the DNLS Hartree orbit trace on the 14-mode multiplet:

$$\frac{\text{Tr } K_{\mathcal{O}_1}}{\text{Tr } K_{\mathcal{O}_0}} = \frac{3}{2}, \quad (39)$$

exact in rational arithmetic. The DNLS Hartree kernel is

$$K_{ab} = \frac{1}{N} \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad (40)$$

with ϕ_a the unnormalized cosine mode wavefunction. Direct numerical verification gives diagonal orbit traces $(2, 3, 2)$ on $(\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2)$.

The mass map reproduces the twelve charged fermion masses at RMS 0.0449 dex ($\sim 11\%$ accuracy) with zero free parameters.

7 Level 5 — Emergent physics

7.1 Flavor: the CKM matrix

Theorem 12 (Cabibbo angle). *The Cabibbo angle satisfies*

$$\sin \theta_{12} = \frac{\pi}{2(L-1)} = \frac{\pi}{14} = 0.2243994752\dots \quad (41)$$

The Wolfenstein parameters are

$$\lambda = \frac{\pi}{14}, \quad A = \frac{d}{d+1}(1 + \delta_{\text{edge}}), \quad \sqrt{\rho^2 + \bar{\eta}^2} = \frac{d+1}{d(d-1)} = \frac{5}{12}, \quad (42)$$

with $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$.

The nine CKM elements partition into three $\mathbb{Z}_3$ -cyclic classes, each satisfying a daughter relation $|V_a| = c_k |V_b| |V_c|$:

$$c_0 = 1, \quad c_1 = \frac{N}{N + (d-1)^d} = \frac{64}{145}, \quad c_2 = w_d \left(1 + \frac{m_u}{m_c} \cdot \frac{N-14}{N-13} \right). \quad (43)$$

Theorem 13 (Anti-daughter coefficients). *The anti-daughter coefficients are*

$$c_1^{\text{anti}} = 1 - \frac{1}{|\mathcal{O}_2|^2} = \frac{15}{16}, \quad c_2^{\text{anti}} = \frac{|\mathcal{O}_2|}{(d-1)^2} = \frac{4}{9}, \quad (44)$$

exact rational identities in the D_4 orbit sizes.

Theorem 14 (Generation phase). *The generation phase is*

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi = \frac{3}{5}\pi = 0.6\pi. \quad (45)$$

7.2 Flavor: the PMNS matrix

The three neutrino mass eigenstates carry the $N/4$ spectrum $(60, 60, 58)$, derived from the four spin structures of the torus with Dirac indices $\pm 1/4$:

$$\left(\frac{N}{4}\right)_g = N - d - 2 \left\lfloor \frac{1 - 4 \text{index}(g)}{2} \right\rfloor. \quad (46)$$

The three non-vacuum spin structures give the labels $(60, 60, 58)$.

With the CKM-analog weights $w_\nu = 3/10$, $w_\ell = 7/10$ and generation phase $\phi_{\text{gen}} = 0.6\pi$, the PMNS magnitudes are reproduced at RMS 0.035. The three mixing angles are $\theta_{12} = 34.2$, $\theta_{23} = 47.9$, $\theta_{13} = 8.5$; the Dirac CP phase is $\delta_{CP} \approx 197$.

Theorem 15 (Majorana phases). *The neutrino mass matrix $M_\nu[i, j] = s_i s_j \sqrt{m_i m_j}$ is real symmetric; its Takagi factorization gives Majorana phases*

$$\alpha_1 = \pi, \quad \alpha_2 = 0 \pmod{\pi}. \quad (47)$$

The Majorana sector of HCSM is CP-conserving. All CP violation in the lepton sector is Dirac, sourced by δ_{CP} .

No HCSM-native complexification — winding bilinear, loop complex time, or their combination — produces Majorana phases that are rational multiples of ϕ_{gen} or $\pi/14$. The zero-state sign structure is the only source, and it gives $(\pi, 0)$.

7.3 Higgs sector

Theorem 16 (Top-Yukawa coefficient). *The top Yukawa satisfies*

$$y_t^2 = 1 - \frac{4}{3}\delta_{\text{edge}}, \quad \frac{4}{3} = \frac{|\mathcal{O}_2|}{d-1}. \quad (48)$$

Prediction: $m_t = 172.688398 \text{ GeV}$, matching PDG $172.690 \pm 0.30 \text{ GeV}$ at 0.0053σ .

Theorem 17 (Tree-level quartic coupling). *The tree-level Higgs quartic coupling is*

$$\lambda_{\text{tree}} = \frac{\ln(kL)}{(d-1)\pi^2} = \frac{\ln(kL)}{\dim(\mathfrak{su}(2)_L)\pi^2}. \quad (49)$$

Theorem 18 (Bridge identity). *The quartic coupling and the vacuum stability coefficient satisfy*

$$\lambda_{\text{tree}} = \frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{2(d-1)}\right), \quad \gamma = \sqrt{d\lambda_{\text{tree}}} \left(1 + \frac{\delta_{\text{edge}}}{4(d-1)}\right), \quad (50)$$

verified at 8.4×10^{-8} in λ_{tree} .

Theorem 19 (Electroweak scale as single anchor). *The electroweak scale is*

$$v_{\text{EW}} = M_P \exp(-kL_{\text{phys}}). \quad (51)$$

M_P is the only dimensionful anchor of the framework.

7.4 Gravity

The graviton is the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, consisting of $54B_1 \oplus 54B_2 = 108$ modes. It is exactly massless at all orders by D_4 Schur orthogonality. The Newton constant is

$$G_N = \frac{25}{8\pi M_P^2} = \frac{(d+1)^2}{8\pi M_P^2}. \quad (52)$$

Theorem 20 (Spectral origin of the factor 25). *The factor $25 = (d+1)^2$ in the Newton constant arises as the trace of the bilinear form on the hidden B_1 isotypic sector:*

$$\text{Tr}_{V_{B_1} \otimes V_{B_1}} \mathbb{I} = (\dim V_{B_1})^2 = (d+1)^2 = 25. \quad (53)$$

8 Level 6 — Cosmology

8.1 The kL crossover

The warp factor is fixed by the mass-map self-consistency condition:

$$kL_{\text{bare}} = \frac{(d-1)N}{d+1} = \frac{192}{5} = 38.4 \text{ exact}, \quad kL_{\text{phys}} = 38.442527, \quad (54)$$

matching $\ln(M_P/v_{\text{EW}}) = 38.442527$ to 0.004%.

Theorem 21 (kL correction coefficient). *The kL correction is*

$$kL_{\text{phys}} - kL_{\text{bare}} = \frac{|D_4| - 1}{2} \cdot \delta_{\text{edge}} = \frac{7}{2} \delta_{\text{edge}}. \quad (55)$$

Precision test: observed ratio $(kL_{\text{phys}} - kL_{\text{bare}})/\delta_{\text{edge}} = 3.50216 \dots$; M_P -limited budget on the ratio is 0.04116; residual/budget = 0.053. Consistent with exactness at 5% of the M_P budget.

8.2 Dark energy and the DME entropy-well normalization

Theorem 22 (Dark energy density).

$$\rho_{\text{DE}} = \frac{|F|}{|B|} m_\nu^4 = \frac{d+1}{2} m_\nu^4 = 2.4809 \times 10^{-47} \text{ GeV}^4, \quad (56)$$

with $m_\nu = 1.7749 \text{ meV}$, matching observation to 0.76%. The equation of state is $w = -1$ exactly from topological Casimir.

Theorem 23 (DME entropy-well normalization). *The DME entropy-well normalization is*

$$S_0 = v_{\text{EW}}^2 \cdot \frac{2d-1}{|F| \cdot |B|} \cdot \left(1 + \frac{\delta_{\text{edge}}}{|F| - |B|}\right) = v_{\text{EW}}^2 \cdot \frac{7}{40} \left(1 + \frac{\delta_{\text{edge}}}{6}\right) = 10630.722 \text{ GeV}^2, \quad (57)$$

matching the previously implied value 10630.696 GeV^2 at relative residual 2.44×10^{-6} , i.e. 18.8% of the M_P precision budget.

Corollary 24 (Single anchor). *HCSM retains its single-anchor structure. M_P is the only dimensionful input. S_0 is derived from $v_{\text{EW}}^2 = M_P^2 \exp(-2kL_{\text{phys}})$ and HCSM-native dimensionless factors.*

8.3 Vacuum stability

The stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188 \dots \quad (58)$$

with

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}), \quad \text{AGM}(1, \sqrt{2}) = \frac{1}{\sqrt{2}\eta(i)^2}, \quad \eta(i) = \frac{\Gamma(1/4)}{2\pi^{3/4}}. \quad (59)$$

The self-dual square torus has modular parameter $\tau = i$; the Chowla–Selberg formula gives $\eta(i)$. The stability theorem is $S''(0)/S_0 = -2\gamma < 0$; the equilibrium $D = 4$ is a strict local maximum of the vacuum entropy.

9 Session-Level Theorem Inventory

The following theorems were established in the September 2026 session and are integrated throughout this paper:

Theorem	Location
$\mathcal{O}_2 = \text{Fix}(M)$; $\mathbb{Z}_2 \times \mathbb{Z}_2$ fixed points = $\{W, \text{top}\}$	§5.7
Top-Yukawa $4/3 = \mathcal{O}_2 /(d-1)$; $m_t = 172.688$ GeV	§7.3
$\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$	§7.3
Bridge identity at 8.4×10^{-8}	§7.3
Hartree orbit trace $(2, 3, 2)$, ratio $3/2$ exact	§6.2
$T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma)$	§5.9
$\sin \theta_{\text{Cabibbo}} = \pi/14$	§7.1
$v_{\text{EW}} = M_P \exp(-kL_{\text{phys}})$ single anchor	§7.3
Spin-structure formula for $(60, 60, 58)$	§7.2
$\phi_{\text{gen}} = (F - B)/ F \cdot \pi = 0.6\pi$	§7.1
Anti-daughter $15/16, 4/9$ from orbit sizes	§7.1
kL correction $7/2 \cdot \delta_{\text{edge}}$ from $ D_4 - 1$	§8.1
$S_0 = v_{\text{EW}}^2 \cdot 7/40 \cdot (1 + \delta_{\text{edge}}/6)$	§8.2
Majorana phases $(\pi, 0)$	§7.2

All rank blockers of the framework’s derivation program are now closed:

Rank	Blocker	Status
1	HCSM-native γ	Closed (HCSM-54)
2	kL bare	Closed (HCSM-51)
3	Charge bijection	Closed (HCSM-24)
4	$N = d \cdot 2^d$	Closed (HCSM-01)
5	$(d-1)/2$ from DNLS	Closed (HCSM-19)
6	3D irrep for CKM	Closed by reframing (HCSM-34)
7	Loop-pass duration	Closed (HCSM-21)

10 Reproducibility

All numerical claims in the HCSM paper stack are verified at machine precision in IEEE-754 double precision. The verification scripts use only standard libraries (NumPy, SciPy, SymPy, mpmath) and construct the objects they need from HCSM-native definitions.

The complete HCSM paper stack, the verification scripts, and the LaTeX and TXT source of each paper are available at

<https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM->

11 Conclusion

This paper has established the foundation of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by two independent anchors (photon helicity for $d = 4$; equal entropy spacing for $L = 8$) and a derived third anchor (the topology of the Hodge–Dirac kernel). The substrate carries the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. The processor T and its five-layer loop are the machine’s operation. The output structures — charge, mass, flavor, gravity, cosmology — are the residue after mirror cancellation.

The framework’s disclosed-input set consists of three discrete inputs (Standard Model content; physical identifications; identification thesis) and one dimensionful anchor (the Planck mass M_P). Every other quantity is derived from the substrate. All seven rank blockers are closed.

The framework’s claim is that the Standard Model is what the substrate looks like when you observe it. The Hodge complex is the machine. Physics is what the machine cannot cancel.

References

- [1] S. Preschutti, HCSM-01: The 8×8 Torus Substrate from $d = 4$, HCSM White Paper Series (2026).
- [2] S. Preschutti, HCSM-02: The Hodge Complex on Λ , HCSM White Paper Series (2026).
- [3] S. Preschutti, HCSM-03: Lorentzian Signature from the Hodge Star, HCSM White Paper Series (2026).
- [4] S. Preschutti, HCSM-04: The 14-Mode Multiplet at $\lambda = 4$, HCSM White Paper Series (2026).
- [5] S. Preschutti, HCSM-06: Sector-Coarse Correspondence, HCSM White Paper Series (2026).
- [6] S. Preschutti, HCSM-13: The Processor T and the F/B Split, HCSM White Paper Series (2026).
- [7] S. Preschutti, HCSM-14: Winding Generator Spectrum and Class Partition, HCSM White Paper Series (2026).
- [8] S. Preschutti, HCSM-17: Loop Operator Cartan Rotation, HCSM White Paper Series (2026).
- [9] S. Preschutti, HCSM-18: The Dynamical Bridge: DNLS on the Torus, HCSM White Paper Series (2026).

- [10] S. Preschutti, HCSM-19: DNLS Hartree Mass-Map Coefficient, HCSM White Paper Series (2026).
- [11] S. Preschutti, HCSM-21: The Time Theorem, HCSM White Paper Series (2026).
- [12] S. Preschutti, HCSM-22: The Zero State, HCSM White Paper Series (2026).
- [13] S. Preschutti, HCSM-24: The Charge Theorem, HCSM White Paper Series (2026).
- [14] S. Preschutti, HCSM-33: Mass Map, HCSM White Paper Series (2026).
- [15] S. Preschutti, HCSM-34: The CKM Matrix, HCSM White Paper Series (2026).
- [16] S. Preschutti, HCSM-35: The Daughter Structure of the CKM, HCSM White Paper Series (2026).
- [17] S. Preschutti, HCSM-36: The PMNS Matrix, HCSM White Paper Series (2026).
- [18] S. Preschutti, HCSM-40: The Higgs Mechanism, HCSM White Paper Series (2026).
- [19] S. Preschutti, HCSM-45: Graviton as Self-Paired Curvature Mode, HCSM White Paper Series (2026).
- [20] S. Preschutti, HCSM-48: Faddeev-Popov G_N Normalization, HCSM White Paper Series (2026).
- [21] S. Preschutti, HCSM-51: The kL Crossover, HCSM White Paper Series (2026).
- [22] S. Preschutti, HCSM-53: Dark Energy Density and the DME Entropy-Well Normalization, HCSM White Paper Series (2026).
- [23] S. Preschutti, HCSM-54: Vacuum Stability and the Stability Coefficient, HCSM White Paper Series (2026).
- [24] Particle Data Group, Phys. Rev. D **110**, 030001 (2024).